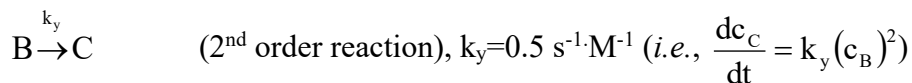
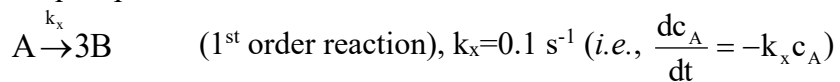


PROBLEM SET 10

Due Apr 21, 2023 @ 8:30 AM via Moodle and Gradescope. Attach screenshots of excel and Matlab portions to the Moodle submission. You will submit your Excel assignments as a single Workbook that contains multiple worksheets (aka sheets, or sheet tabs), as specified below, and Matlab assignment as “.m” files via Moodle. Save your work (*i.e.*, single Excel workbook) as an “Excel Macro-Enabled Workbook” file type (extension .xlsm). Make sure you start a blank workbook on your own computer.

Problem 1 (15 points total)

In HW#9 Problem 1, we asked you to consider the following chemical reactions that take place in a liquid-phase batch reactor of constant volume V.



The reactor is initially charged with a solution of A at a concentration of 0.1 mol/L. No B and C are present at the beginning of the process. Write an “m-file” that enables you to find the concentrations of A, B, and C on the time interval $t \in [0, 80]$ seconds by solving the differential equations (e.g., $dc_A/dt =$, $dc_B/dt =$, and $dc_C/dt =$) with the Matlab function called ODE45. You do so in two steps:

In the first step, you create an “m-file” called **ODE_reaction_ABC**, which defines the differential equations (e.g., dc/dt) that will be used for the ODE45 function. Follow the example we demonstrated in class to create such a function.

In the second step, you write an “m-file”, called **Problem_1**, which uses the ODE_reaction_ABC “m-file” as an input to the ODE45 function. Somewhere in this code you will use:

```
[t,c] = ode45(@ODE_reaction_ABC,tspan,c0);
```

Before you do so, you have to specify what tspan is and what c0 is. Follow the example we showed in class to define tspan and c0, then you call the ODE45 function. Note that c0 in this case will be a vector (not a scalar or single number). So let us help by saying that the definition of c0 is:

```
c0 = [0.1;0;0];
```

Finally, you plot c vs. t. You can do so by typing (in your **Problem_1** “m-file”):

```
plot(t,c);
```

Label the time and charge axes on the plot as “time (s)” and “concentration (mol/L)”, respectively, by using xlabel and ylabel commands written within your **Problem_1** “m-file”.

Problem 2 (10 points total)

Create a worksheet called Problem 1. Record a Macro (called **Problem_2**) that swaps the content of the cells A1:A5 with that of the cells B1:B5. Use data given in **Table 1** to demonstrate this functionality. You can assume that nothing else is written in the worksheet so that you can use any place in the worksheet to do any formatting operation.

	A	B
1	2	4.4
2	4	7.1
3	-0.3	2
4	1.7	9
5	5.5	3

Table 1

↔

	A	B
1	4.4	2
2	7.1	4
3	2	-0.3
4	9	1.7
5	3	5.5

Problem 3 (15 points total)

Create a worksheet called Problem 2. Write a VBA program (called **Problem_3**) that calculates a factorial of a number (*i.e.*, $n!$). Assume that the input is a positive integer $\in (1,7)$ that is stored in cell A1 and you return your result in cell B1.

Problem 4 (15 points total)

You are given a series of numbers in cells A1:A10 (see **Table 2**). Create a worksheet called Problem 4. Write a program (called **Problem_4**) in VBA Excel that scans through the entire series and for evaluates whether each number is even, odd, or zero. The corresponding cells in the B column will thus contain text that reads EVEN, ODD, or ZERO (zero, *i.e.*, “0” is not an even number – despite of what some math teachers may have told you in the past. Zero is simply a special number in mathematics, which ancient mathematicians already recognized a very long time ago¹). For instance, the cell B1 will contain the word ODD, the cell B2 will contain the word EVEN, *etc.* In this assignment you know that you work with exactly 10 numbers. [Hint: How do you know that a number is even? Because it can be divided by 2 giving no residual (*i.e.*, no decimal points). There are multiple ways to solve this problem. One thing you may find useful is a function called INT(number), which returns an integer portion of a number. Hence INT(3.5) returns a value of 3 while INT(2) returns a value of 2. You may also use the following syntax: INT(5/2) returns 2 while 5/2 is 2.5. A very useful function is MOD; it returns the remainder of a division (see, for instance reference 2). Thus the operation 7 MOD 2 returns 1 (that is 7 is not dividable by 2 directly, and it has a residual of 1) while 6 MOD 2 returns 0 (*i.e.*, no residual). There are other functions within Excel that you may use to determine if a number of odd or even... We hope we have not confused you with our hints ☺].

Table 2

	A	B
1	3	
2	6	
3	11	
4	0	
5	9	
6	7	
7	-3	
8	-22	
9	4	
10	5	

Problem 5 (15 points total)

Consider a variation of Problem 4 in this problem set whereby you do not know that you have 10 numbers exactly. You have to somehow make Excel recognize that it should stop when a cell does not contain any number to evaluate (see the example in Excel we created during the class). Create

¹ <https://en.wikipedia.org/wiki/0>

² <http://www.excel-easy.com/vba/examples/mod-operator.html>

a worksheet called Problem 5. Write a program called **Problem_5** that scans through the entire series of numbers in column A and for each number evaluates whether the value is even, odd, or zero. In this case, assume that you do not know that total number of entries in the column A in the spreadsheet (*i.e.*, you do not know that you have only 10 numbers to evaluate). You will simply stop the evaluation when the program encounters an empty cell in column A.

Problem 6 (15 points total)

In HW#9, Problem #3 we asked you to evaluate the 1st derivatives of y with respect to x (*i.e.*, y') from a given data set using the “forward”, “backward” and “central” methods. In this problem you are tasked to do the same by writing a VBA code. The data set is the same as in HW#9, Problem #3. Create a worksheet called Problem 6. Then write a program called **Problem_6** that calculates the 1st derivatives from the given data set using the “forward”, “backward” and “central” methods and stores those in columns D-F (see formatting example in **Figure 1**).

	A	B	C	D	E	F
1	x	y		y'_forward	y'_backward	y'_central
2	0	2				
3	0.05	2.128125				
4	0.1	2.215				
5	0.15	2.264375				
6	0.2	2.28				
7	0.25	2.265625				
8	0.3	2.225				
9	0.35	2.161875				
10	0.4	2.08				
11	0.45	1.983125				
12	0.5	1.875				
13	0.55	1.759375				
14	0.6	1.64				
15	0.65	1.520625				
16	0.7	1.405				
17	0.75	1.296875				
18	0.8	1.2				
19	0.85	1.118125				
20	0.9	1.055				
21	0.95	1.014375				
22	1	1				
23	1.05	1.015625				
24	1.1	1.065				
25	1.15	1.151875				
26	1.2	1.28				
27	1.25	1.453125				
28	1.3	1.675				
29	1.35	1.949375				
30	1.4	2.28				
31	1.45	2.670625				
32	1.5	3.125				
33	1.55	3.646875				
34	1.6	4.24				
35	1.65	4.908125				
36	1.7	5.655				
37	1.75	6.484375				
38	1.8	7.4				
39	1.85	8.405625				
40	1.9	9.505				
41	1.95	10.70188				
42	2	12				

	A	B	C	D	E	F	G
1	x	y			trapezoid	Romberg	Simpson
2	0	2					
3	0.05	2.128125		integral=			
4	0.1	2.215					
5	0.15	2.264375					
6	0.2	2.28					
7	0.25	2.265625					
8	0.3	2.225					
9	0.35	2.161875					
10	0.4	2.08					
11	0.45	1.983125					
12	0.5	1.875					
13	0.55	1.759375					
14	0.6	1.64					
15	0.65	1.520625					
16	0.7	1.405					
17	0.75	1.296875					
18	0.8	1.2					
19	0.85	1.118125					
20	0.9	1.055					
21	0.95	1.014375					
22	1	1					
23	1.05	1.015625					
24	1.1	1.065					
25	1.15	1.151875					
26	1.2	1.28					
27	1.25	1.453125					
28	1.3	1.675					
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33	1.55	3.646875					
34	1.6	4.24					
35	1.65	4.908125					
36	1.7	5.655					
37	1.75	6.484375					
38	1.8	7.4					
39	1.85	8.405625					
40	1.9	9.505					
41	1.95	10.70188					
42	2	12					

Figure 1. (Left column) Format of data sheet for evaluating 1st derivatives of the data given in columns A and B (A is the independent (*i.e.*, x) variable, B is the dependent (*i.e.*, y) variable). The

results will be written in columns B (for “forward” method), C (for “backward” method) and D (for “central” method). (Right column) Format of data sheet for evaluating the integral of the data given in columns A and B (A is the independent (*i.e.*, x) variable, B is the dependent (*i.e.*, y) variable) between $x=1$ and $x=2$. Write the answers for the trapezoidal, Romberg, and Simpson integrations in cells E3, F3, and G3, respectively.

Problem 7 (15 points total)

In HW#9, Problem #3 we asked you to find the approximate integral of the data in the attached Excel File on the interval $x \in [1, 2]$ using the i) trapezoidal rule; ii) Romberg integration; and iii) Simpson 1/3 rule. In this problem you are tasked to do the same by writing a VBA code. Create a worksheet called Problem 7. Then write a program called **Problem_7** that calculates the integral of the data set on the above interval using the three aforementioned methods (see the formatting example in **Figure 1**).